Oligonucleotide-Induced CNS Toxicity: A Technical White Paper

Mechanisms, Clinical Case Studies, Mitigation, and the Knowledge Map of a Fast-Moving Field

Compiled May 22, 2026 — for technical/translational use

TL;DR

- **CNS oligonucleotide toxicity is not one phenomenon but at least four mechanistically distinct ones**: (1) acute biophysical divalent-cation chelation and CSF ionic disturbance driving seizures and motor phenotypes within seconds–hours; (2) late-onset (days–weeks) neuronal apoptosis driven by hybridization-independent PS-ASO protein binding (paraspeckle proteins P54nrb/PSF → nucleolar stress → p53 → caspase activation); (3) hybridization-dependent off-target RNA cleavage by RNase H-recruiting gapmers; and (4) sequence/CpG-driven innate immune activation (TLR9/TLR3/TLR7-8) in microglia and astrocytes. Delivery-related injuries (aseptic meningitis, hydrocephalus, intracranial hypertension, post-LP syndrome) sit on top of these biochemical mechanisms.
- **Clinical reality 2016–2026**: two intrathecal CNS ASOs are approved (nusinersen 2016; tofersen 2023); BIIB080/MAPTRx (tau, Alzheimer's), ION582 (UBE3A-AS, Angelman), zorevunersen/STK-001 (SCN1A, Dravet) and WVE-003 (HTT SNP3, Huntington's) have advanced to pivotal stages. The pivotal cautionary tales are tominersen (Roche GENERATION HD1, halted 2021 for futility with elevated NfL and ventricular volume), WVE-120101/120102 (Wave PRECISION-HD, discontinued 2021 for inefficacy), and branaplam (Novartis VIBRANT-HD, halted 2022 for peripheral neuropathy at 85.7% incidence). PubMed Central Nature Tofersen carries a documented ~7% rate of serious neurological AEs (myelitis, radiculitis, aseptic meningitis, intracranial hypertension/papilledema). PubMed Wiley Online Library
- **Mitigation is now a real discipline, not a hope**: add Ca^{2+}/Mg^{2+} to the formulation (Khvorova, Mol Ther Nucleic Acids 2024); PubMed Central avoid phosphate buffers (Moazami, Mol Ther 2024); nih reduce PS content with mixed PS/PO backbones; Cell Press use sequence-screening algorithms (Hagedorn, Nucleic Acid Ther 2022); use site-specific 5′-cyclopropylene (5′-CP) modifications to abolish paraspeckle-protein-mediated late-onset toxicity (Yamaguchi et al., Mol Ther Nucleic Acids 2025); ScienceDirect use Gap2-OMe to suppress hybridization-independent protein binding (Crooke lab); monitor NfL/GFAP/CSF

protein/ventricular volume as early translational safety biomarkers.

Key Findings

1. **The negatively charged phosphate backbone is itself a toxin at high CSF concentrations.** Each PS/PO bond can chelate Ca^{2+}/Mg^{2+} stoichiometrically; intrathecal/ICV ASO concentrations of 1–4 mM exceed CSF Ca^{2+} (1.3–1.4 mM), producing local divalent cation depletion PubMed Central and seizures within seconds of injection in awake mice. eScholarship Adding excess Ca^{2+}/Mg^{2+} to the formulation prevents seizures without compromising distribution or efficacy PubMed Central (Miller et al., Mol Ther Nucleic Acids 2024).

2. **Sequence matters as much as chemistry.** ~40% of LNA-modified fully-PS ASOs in a 148-compound Roche/BMS screen produced moderate-to-severe acute tolerability findings 1 h after ICV — exactly Hagedorn's phrasing: "about 40% of the ASOs evaluated have moderate to severe acute tolerability observations that preclude further development." Liebert Pub G-richness near the 3′ end is the dominant predictive feature; A-rich sequences are protective. Liebert Pub The Hagedorn calcium-oscillation in-vitro assay correlates with mouse acute tolerability Liebert Pub and can pre-screen out ~half of toxic candidates before in vivo testing.

3. **Late-onset neurotoxicity is mechanistically separable from acute toxicity.** Mislocalization of paraspeckle proteins P54nrb (NONO) and PSF (SFPQ) into nucleoli, nih nucleolar fragmentation, P21/p53 induction, and caspase-mediated apoptosis occur over days–weeks after ICV/IT in rodents. The Yamaguchi/Obika (Osaka) 2025 model shows "hypoactivity in motor function and consciousness persisting for six days or more after dosing." Site-specific 5′-CP at gap positions abolishes this phenotype while preserving knockdown; 2′-OMe at the same position can paradoxically worsen it. nih

4. **Tominersen exemplifies a chemistry- and dose-frequency-dependent inflammatory/CSF-dynamics adverse profile.** Q8W 120 mg yielded ~46% CSF mHTT lowering VJNeurology but increased CSF NfL, International Parkinson and Moveme CSF white-cell count, CSF protein, and ventricular volume — without commensurate parenchymal atrophy New England Journal of Medicine — and produced 3 hydrocephalus cases VJNeurology and clinical worsening vs placebo. GENERATION HD2 is continuing at 100 mg Q16W only after an April 2025 iDMC interim found no safety concerns.

5. **Branaplam shows that toxicity even of a non-oligonucleotide splice modulator hits via the same general "nucleolar stress → p53 → apoptosis" pathway** that Crooke and the Yamaguchi/Obika groups invoke for toxic PS-ASOs — 85.7% peripheral neuropathy in the 56 mg arm of VIBRANT-HD. Nature

6. **The "platform safety" approach now favors mixed PS/PO backbones, MOE wings on**

gapmers, conservative G-content, in silico off-target screening, calcium-containing aCSF/HEPES (not phosphate) buffers, and CSF biomarker monitoring (NfL, GFAP, neuroinflammation panel) from first-in-human. Divalent siRNA (Atalanta) and antibody-oligonucleotide conjugates (Avidity/Dyne/Denali via TfR1) are the leading next-generation delivery platforms.

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1. Introduction & Background

1.1 What "CNS toxicity" means in the oligonucleotide context

For oligonucleotide therapeutics delivered into cerebrospinal fluid (CSF) by intrathecal (IT) lumbar puncture or intracerebroventricular (ICV) injection, CNS toxicity comprises a stratified set of adverse outcomes:

- **Procedural** — post-LP syndrome, headache, back pain, aseptic meningitis from instrumentation;

- **Acute biophysical** — seizures, ataxia, tremor, hypoactivity within seconds–hours of dosing, caused largely by ion chelation and local CSF disequilibrium;
- **Subacute neuroinflammatory** — CSF pleocytosis, CSF protein elevation, microglial/astrocyte activation, NfL elevation, ventricular volume change occurring over days–weeks;
- **Late-onset neurodegenerative** — neuronal apoptosis via hybridization-independent protein binding occurring days–weeks–months after dosing;
- **Hybridization-dependent off-target** — unintended cleavage of partially complementary mRNAs by RNase H-recruiting gapmers, with downstream consequences if essential genes are hit;
- **On-target** — over-suppression of the intended target (e.g., total huntingtin lowering, MAPT loss-of-function).

These categories are not mutually exclusive. For tominersen, all of (a) procedural, (b) subacute inflammatory and (c) on-target lowering hypotheses are still in play.

1.2 Why CNS-directed oligonucleotides are a major therapeutic area

Roughly one in three monogenic diseases with brain phenotypes has an mRNA target that a properly designed ASO or siRNA could engage. Approved or late-stage indications include:

- Spinal muscular atrophy (nusinersen / Spinraza, approved 2016)
- SOD1-amyotrophic lateral sclerosis (tofersen / Qalsody, approved 2023)
- Huntington’s disease (tominersen, WVE-003, ATL-101 in development)
- Alzheimer’s disease (BIIB080 tau ASO, Phase 2 CELIA, NCT05399888)
- Angelman syndrome (ION582, Phase 3 REVEAL, NCT06914609)
- Dravet syndrome (zorevunersen / STK-001, Phase 3 EMPEROR, NCT06872125)
- FUS-ALS (ulefnersen / jacifusen / ION363, Phase 3 FUSION, NCT04768972)
- KCNT1-related epilepsy (Atalanta ATL-201 di-siRNA, IND-stage)

Disease eligibility is broadening because chemistry has matured, and durability after a single intrathecal injection of new siRNA scaffolds (Khvorova di-siRNA: months of silencing in NHP) is reframing dosing economics.

1.3 Delivery routes and why systemic delivery fails for naked oligonucleotides

Naked PS-MOE ASOs and siRNAs do not cross the blood-brain barrier PubMed Central in clinically meaningful amounts. They are large (5–14 kDa), polyanionic, and quickly filtered into the kidney. Brain exposure after IV dosing is typically <1% of liver/kidney levels. CSF administration is required:

- **Intrathecal (lumbar puncture)**: clinical standard for adult/older pediatric indications (nusinersen 12 mg or high-dose 28/50 mg per FDA 2024 label; tofersen 100 mg; tominersen 120 mg). Distribution gradient is cervical-low and CSF-to-parenchyma penetration depends on bulk flow and trans-ependymal transport.
- **Intracerebroventricular**: preclinical standard in mouse and a clinical option in pediatric cases via Ommaya reservoir.
- **Systemic GalNAc-conjugates** are the dominant approach for hepatic targets (vutrisiran, inclisiran, eplontersen) but do not deliver to CNS without an active shuttle.

Emerging exception: antibody-oligonucleotide conjugates (AOCs) targeting transferrin receptor (TfR1) deliver via receptor-mediated transcytosis from the systemic circulation, with proof-of-concept in SMA-mouse via the 8D3 anti-TfR antibody (JCI Insight 2022, PMC7614086). nih PubMed Central Denali's Enzymatic Transport Vehicle (ETV) extends the approach.

1.4 Brief history of CNS oligo development

Year	Milestone
1978	Zamecnik & Stephenson — first antisense concept
1998	Fomivirsen — first ASO approved (CMV retinitis, intravitreal)
2013	Mipomersen approved (HoFH, hepatotoxicity ultimately led to discontinuation)
2016	**Nusinersen / Spinraza** — first CNS-delivered ASO (intrathecal, SMN2 splice switching, 2′-MOE/PS)
2016	Eteplirsen approved (DMD, PMO, IV — first PMO)
2019	Patisiran approved (hATTR-PN, LNP-delivered siRNA)
2020	Givosiran, then GalNAc-siRNAs proliferate
2021	**Tominersen Phase 3 GENERATION HD1 halted for futility/AEs (March 2021)**
2021	Wave PRECISION-HD discontinued
2022	Branaplam VIBRANT-HD halted (peripheral neuropathy)
2023	**Tofersen / Qalsody approved (SOD1-ALS) — first NfL-as-surrogate accelerated approval for ALS**

2023	Eplontersen approved (ATTRv-PN)
2024	Miller/Khvorova — Ca^{2+}/Mg^{2+} formulation rescue (Mol Ther Nucleic Acids)
2024	Moazami/Watts — definitive PS-content/buffer dependency of motor phenotypes (Mol Ther)
2025	Yamaguchi et al. — 5′-CP rescues late-onset neurotoxicity (Mol Ther Nucleic Acids)
2025	Atalanta closes $97M Series B; ATL-201 (KCNT1) and ATL-101 (HTT) di-siRNAs heading to IND
2026	Zorevunersen NEJM publication; EMPEROR Phase 3 enrolling

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2. Mechanisms of CNS Oligonucleotide Toxicity

2.1 Acute toxicity — divalent cation chelation and CSF ionic disturbance

The Khvorova/Aronin mechanism (Miller, Paquette, Barker et al., Molecular Therapy Nucleic Acids 2024; DOI 10.1016/j.omtn.2024.102359, PMC11185713):

Highly concentrated oligonucleotides delivered to CSF in awake mice (instrumented with EEG/EMG) induce **seizures within seconds** of injection. PubMed Central By ion chromatography, both siRNAs and ASOs tightly bind Ca^{2+} and Mg^{2+} up to **molar equivalents of their PO/PS bonds** — independent of structure or PS content. PubMed Central ICV concentrations of 1–4 mM PO/PS thus exceed available CSF Ca^{2+} (1.3–1.4 mM) PubMed Central and Mg^{2+} (~0.7 mM). Adding excess Ca^{2+} alone, Mg^{2+} alone, or both to the formulation **abolishes seizures** with no impact on distribution or efficacy. PubMed Central

The Watts/Moazami refinement (Moazami, Rembetsy-Brown, Sarli et al., Molecular Therapy 2024; DOI 10.1016/j.ymthe.2024.10.024, PMC11638874):

ASOs spanning DNA, 2′-OMe, MOE, and LNA sugar analogs with mixed PS/PO backbones were quantified using a behavioral scoring assay. Key findings:

- **PS-DNA produced the strongest motor phenotype**; 2′-O-substituted RNAs (MOE, OMe) were more tolerable. biorxiv
- Mixed PS/PO backbones were better tolerated than fully PS at equivalent dose. ResearchGate
- **Phosphate-containing buffers (Elliotts B Solution heated, Ca^{2+}/PO_4 mixtures) produced higher phenotype scores than HEPES- or lactate-based buffers** — and can form invisible Ca-phosphate microprecipitates that exacerbate AEs. HEPES-based aCSF is recommended.

- Ca^{2+} pre-formulation also mitigated, but **through a mechanism partially distinct from PS reduction**, nih implying acute toxicity has both ion-chelation and PS-protein-binding components.

The Yokota intracellular-Ca^{2+} mechanism (Jia, Lei Mon, Yang et al., Molecular Therapy Nucleic Acids 2023; DOI 10.1016/j.omtn.2022.12.010, PMC9843489):

In awake mice ICV-injected with toxic LNA gapmers, AMPAR **antagonists potentiated** CNS toxicity while AMPAR agonists and Ca^{2+}-channel activators **mitigated** it. In primary cortical neurons, toxic ASOs **reduce** intracellular free Ca^{2+}. This complements the extracellular-chelation story: depletion of extracellular Ca^{2+} → loss of intracellular Ca^{2+} → loss of normal neuronal excitability balance → paroxysmal seizure phenotype.

Sequence dependence (Hagedorn, Brown, Easton et al., Nucleic Acid Ther 2022; DOI 10.1089/nat.2021.0071, PMC9221153):

Across 148 LNA/PS gapmers ICV-dosed in mice, the authors state: "about 40% of the ASOs evaluated have moderate to severe acute tolerability observations that preclude further development." PubMed Central Sequence features predicting toxicity: **number and 3′-proximity of guanine nucleotides** (G-richness, possibly G-quadruplex tendency); A-content was protective. The Hagedorn calcium-oscillation assay in primary neurons predicted in vivo behavioral scoring. Liebert Pub

2.2 Subacute / late-onset toxicity — paraspeckle protein mislocalization, nucleolar stress, p53

The Crooke mechanism (Vickers, Rahdar, Prakash, Crooke, Nucleic Acids Research 2019, PMC6846478; Crooke et al., Nucleic Acids Research 2020, Vol. 48, No. 10, pp. 5235–5253):

Toxic but not safe PS-ASOs bind more cellular proteins more tightly than non-toxic ones. They cause rapid (within hours of cell entry) **RNase H1-dependent mislocalization of paraspeckle proteins P54nrb (NONO) and PSF (SFPQ) from nuclear paraspeckles to perinucleolar caps and nucleoli**, producing nucleolar stress and fragmentation, up-regulation of P21 mRNA, caspase activation, and apoptosis. Importantly:

- 2′-F-modified PS-ASOs cause sequence-independent proteasomal degradation of P54nrb/PSF.
- Crooke et al. (Nucleic Acids Research 2020) describe that "a simple substitution of a 2′-O-methoxy (OMe) in gap position 2 of toxic PS ASOs resulted in ablation or dramatic reduction in toxicity," with the rescue applying to "nearly all toxic ASOs of all chemical classes studied and all cell lines and organs and species studied to date." This Gap2-OMe rule is the most powerful platform-level rescue described for hybridization-independent toxicity.

**The Yamaguchi/Obika late-onset rescue (Yamaguchi et al., Molecular Therapy Nucleic

Acids 2025; DOI 10.1016/j.omtn.2025.102692, PMC12744863):

Established mouse ICV / rat IT models specifically for **late-onset (≥3 days)** neurotoxicity. Cell Press The corresponding author Takao Yamaguchi (Osaka University) described the model as producing “hypoactivity in motor function and consciousness persisting for six days or more after dosing” even with acute-phase-non-toxic gapmers. Mechanistically:

- p53-regulated transcripts are induced. nih
- Paraspeckle proteins mislocalize to nucleoli. nih
- A novel **5′-cyclopropylene (5′-CP)** modification on gap-region DNA — developed by the Obika lab at Osaka University — at certain positions abolishes late-onset cytotoxicity without compromising knockdown.
- **2′-OMe at the same gap positions can worsen the phenotype** nih — directionally opposite to what the Crooke “Gap2 OMe” rule predicts for acute hepatotoxicity. The two mechanisms are not fully equivalent.

This is one of the most actionable medicinal-chemistry findings of 2025 and is the “2025 5′-CP paper” specifically referenced in the user task brief.

2.3 Hybridization-dependent off-target toxicity

Gapmer ASOs recruit RNase H1, which cleaves as few as 5–6 contiguous DNA:RNA base pairs. Imperfect base pairing to off-target mRNAs containing 5+ contiguous matches can therefore trigger RNase-H-mediated degradation of those off-targets. This mechanism explains:

- The hepatotoxicity of high-affinity LNA/cEt gapmers ScienceDirect (Burel et al., Nucleic Acids Research 2015; Kakiuchi-Kiyota et al., Molecular Therapy Nucleic Acids 2018, PMC5725219).
- Mis-splicing events caused by splice-switching ASOs at imperfectly matched off-targets (Scharner et al., Nucleic Acids Research 2020, PMC6954394).
- Selectivity loss with affinity-increasing modifications: LNA and cEt sharply raise the risk; MOE and morpholinos rarely do.

Standard mitigation: in silico off-target screening (BLAST + match-score), Tm modulation toward physiological window (~50–55 °C), in vitro mouse-fibroblast off-target reduction assays (Kamola et al., Nucleic Acids Res 2015).

2.4 Innate immune activation

ASOs and siRNAs can activate pattern-recognition receptors in CNS-resident immune cells:

- **TLR9** recognizes unmethylated CpG motifs in single-stranded DNA. PS oligonucleotides are stronger TLR9 ligands than PO. Sugar modification of the CpG dinucleotide (e.g., 2′-MOE on

either C or G) sharply reduces TLR9 activation (Yasuhara et al., 2024, PMC11109122). nih
In CNS, intrathecal or intracerebroventricular CpG-ODN causes “loss of neurons, axonal injury in the cerebral cortex, and pronounced microglia activation” (Rosenberger et al., 2014; cited in Frontiers Aging Neurosci 2018, PMC6176466). NCBI

- **TLR3** senses dsRNA — siRNA can engage it.
- **TLR7/TLR8** sense ssRNA — chemical modification (2′-OMe, 2′-F) reduces engagement; this is one reason fully chemically modified siRNAs are essential for in-vivo work.
- **Cytokine signatures**: microglia and astrocytes both produce IL-6, TNF, type I IFN; microglia also produce IL-10/G-CSF/IL-9 anti-inflammatory cytokines that may modulate the response (Olson 2009, PMC3767435). NCBI
- **Complement activation** is generally less prominent in CNS than periphery for naked oligonucleotides, but conjugated or formulated material (lipid nanoparticles, AOCs) can engage complement.

2.5 Delivery-related toxicity

- **Aseptic meningitis** is a documented post-marketing adverse reaction for nusinersen (Spinraza FDA label 2024 update) and reported for ~2% of tofersen-treated patients in VALOR/OLE; mechanism is mixed (procedural + chemistry-driven CSF inflammation).
- **Hydrocephalus** has been reported with nusinersen post-marketing (FDA label) and appeared in 3 patients on tominersen Q8W in GENERATION HD1. VJNeurology
- **Intracranial hypertension and papilledema** account for 4 of the 12 serious neurologic AEs in tofersen clinical trials. Wiley Online Library
- **Myelitis and lumbar radiculitis** also appeared in tofersen (4 of 12 serious neurologic AEs were myelitis; 2 were radiculitis). Wiley Online Library
- **Catheter/pump complications** can arise from Ommaya reservoirs or implanted intrathecal pumps.
- **Hyponatremia and proteinuria** have been seen with nusinersen — likely reflecting systemic ASO exposure after CSF egress; the FDA label requires baseline and pre-dose urine protein monitoring. Spinrazahcp

2.6 Chemistry-specific toxicities

Modification	Acute CNS tolerability	Late-onset neurotoxicity	Off-target risk	Innate immune risk	Notes
PO	Best (minimal				Nuclease-

(phosphodiester)	Ca^{2+} chelation per bond)	Lowest	Lowest	Lowest	labile; rare alone
PS (phosphorothioate)	Worst when fully-PS	Substantial protein binding	Sequence-dependent	TLR9 if CpG	Standard backbone; mix with PO to mitigate
2′-OMe	Moderate	**Can worsen late-onset toxicity at gap positions (Yamaguchi 2025)**	Moderate	Suppresses TLR8	Standard sugar modification
2′-MOE	Good (clinical workhorse)	Lower than DNA	Lower than LNA/cEt	Suppresses TLR9 if at CpG	Nusinersen, tofersen, BIIB080 all 2′-MOE/PS
2′-F	Moderate	**Strong — causes P54nrb/PSF degradation**	Moderate	Suppresses TLR	Useful in siRNAs, risky in gapmers
LNA (BNA)	Poor in gapmers; high hepatotoxicity	Linked to nucleolar stress	High (high Tm → off-target tolerance)	Variable	Powerful but narrow therapeutic index
cEt (constrained ethyl)	Similar to LNA	Similar to LNA	Similar to LNA	—	Ionis next-gen chemistry; tofersen contains cEt-MOE in wings
5′-cyclopropylene (5′-CP)	Comparable to MOE	**Rescues late-onset toxicity (Yamaguchi 2025)**	Sequence-dependent	—	New Obika lab modification, 2025
					Eteplirsen, golodirsen;

PMO / morpholino	Excellent (uncharged)	Minimal	Lowest of all	Lowest	poor CNS penetration without conjugation
PMO + cell-penetrating peptide (PPMO)	Variable	Variable	Lowest	Variable	Peptide-driven tox; SRP-5051 PPMO in DMD trial

The "PS dose" hypothesis: a clear monotonic relationship has been demonstrated between number of PS linkages in an ASO and acute motor phenotype severity in mice (Moazami 2024). All clinically qualified gapmer ASOs now use mixed PS/PO backbones. Cell Press

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3. Clinical Case Studies

3.1 Nusinersen (Spinraza, Biogen/Ionis) — SMN2 splice switcher

- **Chemistry**: 18-mer, fully 2′-MOE / fully PS, splice-switching steric blocker (no RNase H recruitment).
- **Target**: SMN2 intron 7 ISS-N1, promotes exon 7 inclusion → functional SMN protein.
- **Dosing**: 12 mg intrathecal (4 loading doses then Q4M); the 2024 FDA label additionally approves a **high-dose regimen of 50 mg loading + 28 mg maintenance**. Biogen
- **Safety profile (FDA label, accessdata.fda.gov 209531s013s014lbl.pdf 2024)**:
    - Most common AEs: respiratory infection, fever, constipation, headache, vomiting, back pain. Biogen
    - **Class-effect monitoring required**: thrombocytopenia and coagulation abnormalities, renal toxicity (urinary protein), hepatic effects.
    - **Post-marketing**: hydrocephalus, aseptic meningitis, hypersensitivity (angioedema, urticaria), arachnoiditis. FDA
    - Severe hyponatremia in one infant requiring salt supplementation for 14 months. Spinrazahcp
- **Notable**: nusinersen sets the benchmark for "well-tolerated" intrathecal ASO. Its splice-switching mechanism avoids RNase-H-driven off-target risk.

3.2 Tofersen (Qalsody, Biogen/Ionis) — SOD1-ALS gapmer ASO

- **Chemistry**: 2′-MOE/PS gapmer with 5-10-5 architecture; RNase H mechanism.
- **Trial**: VALOR Phase 3, NCT02623699; OLE NCT03070119.
- **Efficacy**: VALOR did not meet primary endpoint (ALSFRS-R decline) at 28 weeks; OLE and combined early-vs-delayed-start analyses showed slowing of decline; **CSF SOD1 protein decreased and serum NfL decreased substantially**, supporting accelerated approval.
- **Approval**: April 2023, FDA accelerated approval **using NfL as a surrogate biomarker** — a landmark for ALS regulatory science.
- **Safety profile (Miller et al., NEJM 2022, DOI 10.1056/NEJMoa2204705; Lovett et al., Muscle Nerve 2025, PMC12060635)**:
    - **CSF pleocytosis in 58% of tofersen-treated vs 6% placebo**; ResearchGate CSF protein increased by ~110 mg/L. PubMed Central
    - Lovett et al. (Muscle Nerve 2025) state verbatim: "Ten participants (approximately 7% of tofersen 100-mg–treated trial participants) experienced a total of 12 serious neurologic AEs—4 of myelitis, 2 of radiculitis, 2 of aseptic meningitis, and 4 of intracranial hypertension (ICH) and/or papilledema." PubMed Wiley Online Library
    - Myelitis in one patient required glucocorticoids + plasma exchange with imaging/symptom resolution by 3 months. New England Journal of Medicine
    - "All serious neurological adverse events were reversible; few led to tofersen discontinuation." PubMed Central
    - German real-world early access (Meyer et al., eClinicalMedicine 2024): "two patients with autoimmune-like myeloradiculitis," confirming the inflammatory signal in routine use.
- **Implication**: tofersen's 7% serious neurologic AE rate is the most quantitatively documented profile of CNS-delivered gapmer toxicity in current clinical use; the inflammatory mechanism is unsettled but PS-binding-mediated paraspeckle perturbation is plausible.

3.3 Tominersen (Roche/Ionis) — non-allele-selective HTT gapmer

- **Chemistry**: 2′-MOE/PS gapmer.
- **Trial**: GENERATION HD1, NCT03761849; **791 manifest HD patients** Neurology Live randomized to placebo, 120 mg Q8W, or 120 mg Q16W intrathecal.
- **Halted March 2021** for futility per iDMC.
- **Key findings (Tabrizi et al., NEJM 2025, DOI 10.1056/NEJMc2300400 and full Phase 3 publication)**:

- 17-month CSF mHTT reductions: **46% Q8W, 29% Q16W**. VJNeurology
- **Q8W arm worsened clinically vs placebo** on cUHDRS; Q16W arm was comparable to placebo. VJNeurology
- **CSF NfL increased** in the Q8W arm; VJNeurology **ventricular volume increased dose-dependently** VJNeurology without matched parenchymal atrophy. Cureffi New England Journal of Medicine
- **3 hydrocephalus cases in Q8W vs none in placebo or Q16W**; VJNeurology SUSARs 1.9% Q8W vs 0.4–0.8%; SAEs 15% Q8W vs 8–11%. Cureffi
- Mechanism: per Roche's own analysis, ventricular increase is "perhaps due to the reduced absorption of CSF, which in turn could be due to an increased white-cell count and protein levels in CSF." New England Journal of Medicine Possible compounding contributors: PS-binding-protein paraspeckle toxicity, on-target HTT lowering effects, and TLR-driven inflammation.

- **GENERATION HD2 (NCT05686551, Roche April 2025 community letter)**: 301 prodromal/early manifest HD patients, originally 60 mg vs 100 mg Q16W vs placebo. Neurology Live **iDMC interim analysis (April 2025): no safety concerns; 60 mg arm discontinued in favor of 100 mg.** Huntington's Disease Society of Am Trial blinded; completion expected 2026. Huntington-disease
- **Implication**: tominersen is the definitive cautionary tale of CNS oligonucleotide tox; both chemistry (PS gapmer) and target biology (huntingtin lowering) remain confounded.

3.4 Wave Life Sciences PRECISION-HD and SELECT-HD

- **WVE-120101 (SNP1, HTT rs362307) and WVE-120102 (SNP2, HTT rs362331)**: stereopure first-generation PS ASOs. Trials NCT03225833 (PRECISION-HD1) and NCT03225846 (PRECISION-HD2). clinicaltrials Discontinued March 2021 — no statistically significant mHTT lowering; PRECISION-HD2 32 mg achieved 9.9% median CSF mHTT reduction (P = 0.74). Neurology Live Reasonable safety but no efficacy.
- **WVE-003 (SNP3 / rs362307 HTT)**: PN-backbone (next-generation phosphoryl-guanidine + stereopure) allele-selective ASO. Trial SELECT-HD, NCT05032196.
  - June 2024 multidose readout (n=16 30 mg Q8W + n=7 placebo): **mean 46% CSF mHTT reduction at week 24, P=0.0007**; Neurology Live **wtHTT preserved or increased**; BioPharma Dive statistically significant correlation between mHTT lowering and caudate-atrophy slowing (R = –0.50, P = 0.047). Neurology Live
  - **No SAEs; ventricular volume in line with natural history; NfL mostly placebo-range.** Wave Life Sciences
  - FDA Orphan Drug Designation; Help 4 HD Internatio receptive to caudate atrophy as

primary registrational endpoint; IND for global Phase 2/3 expected 2H 2025. Wave Life Sciences

- **Implication**: allele-selective design with reduced PS exposure appears to overcome both tominersen's inflammatory and on-target liabilities.

3.5 Branaplam (Novartis LMI070) — splice modulator, parallel cautionary tale

- **Chemistry**: oral small-molecule splice modulator of HTT (and SMN2); not an oligonucleotide.
- **Trial**: VIBRANT-HD, NCT05111249. 21 patients in initial 56 mg weekly cohort.
- **Outcome (Schobel et al., Nature Medicine 2025, DOI 10.1038/s41591-025-04117-4)**:
  - **18 of 21 (85.7%)** showed at least one peripheral neuropathy sign or symptom. PubMed Central Nature
  - First splicing modulator to lower CSF mHTT in HD patients, Nature but **NfL rose** Nature before histological changes; **dose-modeling indicated lower doses would not reach therapeutic mHTT lowering**. PubMed Central
  - Permanently discontinued Nature December 2022. PubMed Central
- **Mechanism (Schur et al., iScience 2025, PMC12059699)**: branaplam activates p53, induces nucleolar stress in iPSC-MNs, and enhances neurotoxic BBC3 expression PubMed Central nih — **the same nucleolar-stress/p53 axis identified for late-onset toxic PS-ASOs by Crooke and Yamaguchi**. This convergence suggests a unifying "nucleolar stress" toxicity pathway across modalities.

3.6 BIIB080 / MAPTRx (Biogen/Ionis) — tau ASO for Alzheimer's

- **Chemistry**: 2′-MOE/PS 18-mer ASO complementary to MAPT pre-mRNA; Nature intrathecal.
- **Phase 1b (Mummery et al., Nature Medicine 2023, DOI 10.1038/s41591-023-02326-3)**: 46 mild-AD patients; doses 10/30/60 mg Q4W or 115 mg Q12W. Nature Per the paper: "BIIB080 was generally well tolerated and was associated with a dose-dependent reduction in CSF t-tau and p-tau181 in the MAD period (56% reduction; 95% CI, 50% to 62%)" with p-tau181 reduced by 51% (95% CI, 38% to 63%) in the higher-dose cohorts; ResearchGate biomarkers continued declining to ~60% below baseline in the 60 mg monthly and 115 mg quarterly groups during the LTE. ALZFORUM Tau PET reduced across all composites. nih Biogen **Generally well tolerated**, no drug-related SAEs.
- **Exploratory analysis (Mummery et al., Nature Aging 2025, DOI 10.1038/s43587-025-01031-9)**: numerical favorable trends on cognitive/functional measures in high-dose groups.

- **Phase 2 CELIA (NCT05399888)**: ~735 patients with MCI or mild AD; Alzdiscovery fully enrolled; data expected 2026. Pharmacy Times
- **Implication**: longest-running and best-tolerated tau-lowering ASO; demonstrates that with careful design, a CNS gapmer-like ASO can avoid both the tominersen inflammatory profile and the late-onset neurotoxicity seen in the Yamaguchi screen.

3.7 Ulefnersen / jacifusen / ION363 (Ionis, licensed Otsuka 2024) — FUS-ALS

- **Chemistry**: 2′-MOE/PS gapmer targeting FUS intron 6. Yuntsg
- **Compassionate-use case series (Shneider et al., Lancet 2025)**: 12 patients, dose escalation 20→120 mg monthly; ScienceDirect **CSF NfL decreased up to ~83% at 6 months**; Oligotherapeutics one P525L patient showed objective functional recovery; Yuntsg post-mortem (n=4) showed wide CNS distribution and near-undetectable FUS protein. ALZFORUM
- **Safety**: "generally well tolerated; no serious side effects related to therapy." ALS News Today Most common possibly-related AE: headache (post-LP). ALS News Today
- **Phase 3 FUSION**: NCT04768972, ongoing under Otsuka.
- **Implication**: large NfL reduction in a fatal disease provides a strong contrast to tominersen's NfL increase, and a strong case for NfL as a target-engagement and disease-modification biomarker.

3.8 Zorevunersen / STK-001 (Stoke Therapeutics/Biogen) — Dravet SCN1A

- **Chemistry**: TANGO splice-switching 2′-MOE/PS ASO; promotes inclusion of productive SCN1A mRNA splice form → upregulates Nav1.1. UCSF
- **Phase 1/2a (MONARCH NCT04442295 / ADMIRAL) and OLE (SWALLOWTAIL, LONGWING)**:
  - 81 patients, >800 doses analyzed. Stock Titan
  - **70 mg cohorts: median convulsive seizure reductions of 85% at 3 months and 74% at 6 months**; Neurology Live **median 58–90% reduction sustained up to 36 months** in OLE. Jobs AC UK
  - Improvements in cognition, behavior (Vineland), motor skills, communication. Stock Titan
- **Safety**:
  - **Most common treatment-related AE: CSF protein elevation in 44% (33/75) of OLE patients** Stock Titan — asymptomatic.

- 1 discontinuation for CSF protein elevation. Stock Titan
- 3 deaths (2 SUDEP, 1 malnutrition) deemed unrelated to drug. Jobs AC UK

- **Phase 3 EMPEROR (NCT06872125)**: 2 loading doses (70 mg Day 1 + Week 8) + 2 maintenance doses (45 mg Week 24 + 40); Biogen ~150 patients. Neurology Live Data expected mid-2027. Stoke Therapeutics
- **Publication**: Laux et al., NEJM 2026 (DOI: 10.1056/NEJMoa2506295). OC Academy

3.9 ION582 (Ionis) — Angelman syndrome UBE3A-ATS

- **Chemistry**: 2′-MOE/PS ASO targeting UBE3A antisense transcript / SNHG14 to derepress paternal UBE3A. Practical Neurology
- **HALOS Phase 1/2a (NCT05127226)**: 51 patients ages 2–50; Ionis **65% improved on Bayley-4 cognition, 65% on fine/gross motor, 70% on receptive communication**, Neurology Live exceeding natural history; Ionis "safe and well tolerated at all dose levels." PR Newswire
- **REVEAL Phase 3 (NCT06914609)**: first patient dosed June 11, 2025; Business Wire amended December 2025 to single 80 mg active arm vs placebo, Veeva ~200 patients. Ionis

3.10 Eteplirsen / golodirsen / casimersen / SRP-5051 — DMD (peripheral, but instructive)

- **Chemistry**: PMO (eteplirsen, golodirsen, casimersen); SRP-5051 is a PPMO with the Pip6a-class cell-penetrating peptide.
- **CNS relevance**: PMOs are uncharged, do not chelate divalent cations, and have minimal innate immune signature — they represent a chemistry-class "control" for what CNS toxicity looks like in the absence of PS backbone. SRP-5051 had renal tubular toxicity in DMD trials likely attributable to the cationic CPP — informative for peptide-conjugate strategies in CNS (e.g., DG9-PMO for SMA, Hammond's Pip6a-PMO for SMN2).

3.11 Inotersen / mipomersen / patisiran / vutrisiran / eplontersen — peripheral precedents

- Mipomersen (apoB) and inotersen (TTR) were both fully-PS 2′-MOE gapmers with substantial hepatotoxicity and (for inotersen) thrombocytopenia; mipomersen was withdrawn, inotersen carries a black-box warning.
- **Patisiran (LNP-siRNA, TTR, 2018) and vutrisiran (GalNAc-siRNA, TTR, 2022, both Alnylam)** established the durability and safety of fully-modified siRNAs systemically — the chemistry foundation that di-siRNAs extend to CNS.

- **Eplontersen / WAINUA (Ionis/AstraZeneca, 2023)**: GalNAc3-conjugated 2′-MOE/PS LICA ASO, sustained ~81% TTR reduction at Week 85; thrombocytopenia rate comparable to placebo (cf. inotersen 13%). PubMed Central Demonstrates that GalNAc-conjugation dramatically reduces dose requirements and thereby ASO-related off-target toxicity — a principle that the field is now trying to translate to CNS via TfR1 conjugates and divalent siRNAs.

3.12 Atalanta ATL-101 and ATL-201 — di-siRNAs (preclinical, IND-stage)

- **ATL-101 (HTT)**: single intrathecal di-siRNA dose produces potent HTT silencing including deep brain regions, 6-month durability, "excellent tolerability." BioSpace Global Genes
- **ATL-201 (KCNT1)**: Andreone et al., Epilepsia 2025 shows near-complete seizure suppression for 6 months in a KCNT1 mouse model. Atalantatx +2
- **Pipeline**: Huntington's, KCNT1 epilepsy, pain, Alzheimer's; Genentech partnership. Atalantatx
- **No human data yet** (no NCT as of mid-2025); IND submissions planned 2025. Atalantatx Business Wire

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4. Preclinical Assessment Methods

4.1 In vitro assays

Assay	Readout	Predictive for	Reference
Hagedorn neuronal Ca^{2+}-oscillation assay	Spontaneous Ca^{2+} wave amplitude/frequency in primary cortical neurons or iPSC-derived cultures	Acute ICV/IT toxicity in mouse	Hagedorn 2022 (PMC9221153)
PS-ASO/P54nrb fluorescence binding	Real-time NLuc complementation reporter for PS-ASO/protein interaction	Hybridization-independent toxicity, late-onset risk	Vickers 2019 (PMC6846478)
MEA (multi-electrode array) for neuronal activity	Spike-rate, network burst, oscillation parameters	Acute neuronal-network disruption	Standard ToxCast platform
iPSC-derived neuronal	LDH, ATP, caspase-3	Late-onset	Yamaguchi 2025

cytotoxicity		apoptotic toxicity	(PMC12744863)
In vitro hybridization off-target Tm	Melting curve to predicted off-target sites	Hybridization-dependent toxicity	Kakiuchi-Kiyota 2018
NEAT1/paraspeckle imaging in HeLa or neurons	Mislocalization of P54nrb/PSF to nucleoli	Toxic vs safe PS-ASO classification	Liang et al., NAR 2019
TLR9 NF-κB reporter in HEK-Blue TLR9	NF-κB reporter induction	Innate immune liability of CpG-containing ASOs	Standard InvivoGen platform
Microglia/astrocyte mixed cultures	Cytokine secretion (IL-6, TNF, IFN-α/β)	Neuroinflammatory liability	Olson 2009 (PMC3767435)

4.2 In vivo models

- **Mouse ICV in conscious or anesthetized animals** with awake EEG/EMG (Khvorova/Aronin, Watts) — gold standard for acute behavioral and electrographic phenotyping. Volume typically 5–10 µL.
- **Mouse acute behavioral scoring (Moazami/Watts behavioral scoring; Hagedorn 0–3 tolerability scale)**: standardized timing/severity scores capturing seizures, ataxia, hyperactivity, immobility.
- **Mouse late-onset model (Yamaguchi 2025)**: rotarod, open-field, foot-fault scoring at 3+ days post-ICV; rat IT for hindlimb weakness/sensory testing.
- **Rat intrathecal (lumbar)**: more translational for human IT delivery; standard CSF biomarker sampling possible.
- **NHP intrathecal**: Cynomolgus or rhesus IT under fluoroscopy; gold standard for IND-enabling biodistribution and safety pharmacology.
- **Sheep large-brain model** (Watts lab Moazami 2024): bridges mouse and NHP for CSF-flow-dependent distribution and safety.

4.3 Behavioral/electrophysiological readouts

- Acute behavioral scoring (Hagedorn 0–3; Watts/Moazami extended 0–5).
- Rotarod, open-field, beam-walking for subacute motor.
- EEG/EMG telemetry (in awake mice using Pinnacle systems; Khvorova 2024 protocol).
- Auditory startle, prepulse inhibition (limbic/sensorimotor gating).

- von Frey filaments and tail-flick for sensory/peripheral neuropathy.

4.4 Biomarkers

- **CSF NfL (Quanterix Simoa)** — universal marker of neuroaxonal injury; central to tofersen approval and to tominersen failure interpretation. Elevations within 1–4 months of dosing are concerning; sustained elevations are predictive of unfavorable outcomes.
- **CSF GFAP** — astrocyte activation marker; rises in tominersen-treated patients.
- **CSF total tau / p-tau181** — useful in AD ASO trials (BIIB080).
- **CSF cell count and protein** — direct readout of meningeal inflammation; tofersen patients have 58% pleocytosis incidence; PubMed Central tominersen patients had elevations correlated with ventricular volume change.
- **Plasma NfL** — convenient longitudinal serum biomarker; correlates with CSF NfL for most CNS diseases.
- **Ventricular volume on MRI** — emerging structural safety endpoint; both tominersen and branaplam used it.
- **DTI/MR spectroscopy** — research-stage; underused in CNS ASO trials.

4.5 Histopathology

Standard nonclinical IND-enabling work-up includes:

- H&E for gross neuronal loss, vascular changes;
- Iba1 (microglia), GFAP (astrocytes), CD68 (activated macrophages);
- NeuN / Fluoro-Jade for neuronal injury/death;
- Cleaved caspase-3 for apoptosis;
- Demyelination (Luxol fast blue, MBP IHC);
- ISH or RNAscope for ASO biodistribution.

4.6 In silico tools

- BLAST + off-target match-score (standard).
- Roche/BMS “G-content + 3′ position” acute neurotoxicity score (Hagedorn 2022).
- Ionis hepatotoxicity machine-learning models.
- Tm prediction tools (OligoAnalyzer, etc.).
- “Sequence-chemistry collusion” ML approaches (chemRxiv 2025-d2bhr).

5. Mitigation Strategies

5.1 Formulation: Ca^{2+}/Mg^{2+} and buffer choice

- **Add Ca^{2+} alone, Mg^{2+} alone, or both at molar equivalents (and ideally excess) of total PO+PS bonds** in the dose volume — prevents acute seizure phenotype without affecting distribution or efficacy (Miller/Khvorova 2024). For a 4 mM ASO solution with 20 PS bonds, ~80 mM divalent cation is the chelation-saturating concentration; the buffer must support this without precipitation.

- **Use HEPES- or lactate-based artificial CSF, not phosphate buffer**: phosphate + Ca^{2+} forms Ca-phosphate microprecipitates even in clear solutions (Moazami 2024). Use Elliotts B only with caution and avoid heat/freeze-thaw cycles.

- **Verify particle-free status by DLS** before injection. The Watts lab established an ultrafiltration buffer-exchange protocol (Amicon cartridges) to achieve calcium-containing, microparticle-free preparations.

5.2 Chemistry: PS content reduction and strategic modifications

- **Reduce PS content via mixed PS/PO backbones**: a PS/PO mixed-backbone gapmer is significantly better tolerated than fully-PS, with no detectable EEG/EMG seizures, modest hyperactivity at most (Khvorova 2024 Figure 7). All clinically qualified gapmer ASOs now use mixed PS/PO.

- **Gap2-OMe (Crooke)**: 2′-OMe at position 2 of the gap region reduces PS-ASO/protein binding. Oxford Academic Crooke et al. (NAR 2020) describe that "a simple substitution of a 2′-O-methoxy (OMe) in gap position 2 of toxic PS ASOs resulted in ablation or dramatic reduction in toxicity," with effects applying to "nearly all toxic ASOs of all chemical classes studied and all cell lines and organs and species studied to date."

- **5′-CP at gap positions (Yamaguchi/Obika 2025)**: site-specific 5′-cyclopropylene rescues late-onset CNS neurotoxicity in mouse ICV and rat IT models without sacrificing knockdown. The first medicinal-chemistry rescue specific to the late-onset phenotype.

- **Avoid 2′-F in gap regions** of CNS ASOs (causes P54nrb/PSF proteasomal degradation). PubMed Central

- **For gapmers, avoid LNA/cEt at maximally toxic positions**; use MOE wings preferentially.

- **G-clamp / aminoethoxyphenoxazine modifications** (Tagami/Yokota 2024) act as putative sigma-1 receptor agonists and reduce acute CNS toxicity while preserving efficacy.

- **Stereopure synthesis (Wave)**: chirally defined PS centers can reduce some sequence-dependent toxicity by selecting Rp/Sp combinations with reduced protein binding.

5.3 Sequence design

- **Avoid G-rich stretches near the 3′ end** (Hagedorn). Liebert Pub
- **Avoid CpG motifs** unless TLR9 activation is therapeutically desired; if CpG present, sugar-modify the C and/or G to abolish TLR9 binding (Yasuhara 2024).
- **Avoid TGC/TCC and other Burdick hepatotoxicity motifs** even in CNS gapmers — they correlate with hybridization-independent toxicity.
- **Run BLAST off-target screens with permissive mismatch settings** (allow 1–2 mismatches in the gap region) to identify potential RNase-H off-targets, especially long pre-mRNAs.

5.4 Dose and regimen

- **Fractionate dosing**: tominersen's Q8W dosing was clearly worse than Q16W. NfL accumulation kinetics may favor longer interdose intervals.
- **Use lowest dose achieving target engagement** (typically guided by CSF biomarker reduction). Compare to tofersen (CSF SOD1 ↓35–55% at 100 mg) and BIIB080 (CSF tau ↓56% per Mummery 2023 MAD, ↓~60% by end of LTE).
- **Dose-escalate in early phase** with adaptive stopping rules tied to NfL trajectory.

5.5 Delivery devices and procedure

- Use ultrasound or fluoroscopy-guided LP to minimize procedural AEs (post-LP syndrome, radiculopathy).
- For chronic dosing, consider intrathecal port systems (e.g., Medstream, SynchroMed) — but recognize device-related complications.
- For pediatric or chronic CNS programs, ICV via Ommaya reservoir can reduce procedural variability.

5.6 Biomarker monitoring

- **Tier 1 (every dose)**: serum NfL, CBC, CMP, urine protein (per Spinraza label), CSF cell count + protein at sentinel timepoints.
- **Tier 2 (quarterly)**: CSF NfL, GFAP, cytokine panel (IL-6, TNF, IFN-α/β); MRI ventricular volume.
- **Tier 3 (annually or per cohort)**: full neuropsychological and electrophysiology workup; PET (tau, amyloid, microglia TSPO) as indication-relevant.

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6. Regulatory & Translational Considerations

6.1 FDA guidance

- **No standalone FDA guidance on oligonucleotide therapeutics exists** as of May 2026, but FDA has hosted multiple workshops (CDER 2017, 2019, 2022) and the agency follows a hybrid of small-molecule and biologic frameworks.
- **ICH S6(R1)** (biotechnology-derived pharmaceuticals) is the closest analog; ICH S7A/S7B safety pharmacology applies.
- **OND Office of Neuroscience** is the primary review division for CNS ASOs.
- **Accelerated approval pathway** is now well-established for CNS ASOs targeting rare/fatal diseases with surrogate biomarkers — tofersen (NfL) is the precedent; WVE-003 (caudate atrophy) and others are pursuing similar pathways.

6.2 EMA considerations

- EMA's Committee for Advanced Therapies (CAT) reviews complex oligonucleotide constructs; PRIME (PRIority MEdicines) designation is available.
- EU label requirements emphasize platelet, hepatic, and renal monitoring for the ASO class.

6.3 ICH guidelines

- ICH S6(R1): biotechnology-derived therapeutics
- ICH S7A: safety pharmacology core battery
- ICH S7B: QT/cardiac
- ICH M3(R2): nonclinical safety studies timing
- ICH S5(R3): reproductive toxicology
- For CNS ASOs, NHP IT is generally required for IND; chronic NHP toxicology with biomarker panels is standard.

6.4 NfL as a regulatory biomarker

The FDA's acceptance of plasma NfL reduction as the surrogate basis for tofersen accelerated approval (April 2023) is the single most important regulatory development for CNS ASOs. It implies that other genetically defined neurodegenerative diseases with robust NfL biomarker characterization may be amenable to similar pathways.

6.5 "Platform" toxicology

For sponsors with multiple ASOs of the same chemistry class (e.g., Ionis's 2′-MOE/PS gapmer platform), the FDA accepts cross-program platform toxicology data; this means that chemistry-class safety signals (e.g., thrombocytopenia, complement activation) need not be characterized de novo for each new ASO. The class-effect labeling for Spinraza (renal, hepatic, platelet) is the model.

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7. Emerging Approaches

7.1 Divalent siRNAs (di-siRNAs)

Two fully-modified asymmetric siRNAs connected by a 3′-3′ linker (Alterman/Khvorova/Aronin, Nature Biotechnology 2019, PMC6879195). A single ICV/IT injection achieves widespread cortical, striatal, and spinal-cord HTT silencing for 6+ months in mice and NHPs, with no detectable toxicity in NHP and minimal off-target effects. Atalanta Therapeutics is commercializing the platform Huntingtonsdiseasenews (ATL-101 HTT; ATL-201 KCNT1 epilepsy; Alzheimer's, pain, additional CNS targets with Genentech). Fierce Biotech

7.2 Antibody-oligonucleotide conjugates (AOCs) for CNS

- **Anti-TfR1 antibody/Fab fragments as BBB shuttles**: Avidity AOC 1001 (DMPK siRNA, peripheral DM1) is the proof-of-concept for systemic AOC delivery; **Denali's ETV platform** explicitly engineered for CNS via BBB transcytosis (DNL310 for Hunter syndrome IDS protein delivery; ASO/siRNA cargos in development).
- **JCI Insight 2022 (PMC7614086)** demonstrated 8D3-anti-TfR-ASO conjugate achieving therapeutic SMN2 splicing in CNS after systemic dosing in mice. nih
- Trade-offs: full IgG retains FcRn recycling/half-life but adds Fc effector risk; Fab fragments penetrate better but require more frequent dosing. Wuxiapptec

7.3 Lipid-conjugated siRNAs for CNS

Brown et al. (Alnylam, Nature Biotechnology 2022): lipid (cholesterol, docosanoic acid) conjugates of fully-modified siRNAs achieve extrahepatic delivery to CNS, lung, muscle. Lipid identity tunes tissue selectivity and PK.

7.4 Peptide-conjugated PMOs (PPMOs) for CNS

- Hammond/Wood lab Pip-series (Pip6a, Pip6f) PMOs enhance CNS delivery after systemic injection but are toxicologically constrained.
- Yokota lab DG9-PMO (PMC10077475) — DG9 from Hph-1 transcription factor — achieves CNS delivery of SMN2 PMO via subcutaneous injection nih with no apparent toxicity in mouse, a notable advance.

- Sarepta SRP-5051 PPMO showed dose-limiting renal toxicity in DMD trials — cautionary for cationic-CPP CNS programs.

7.5 AAV-delivered shRNAs / amiRNAs

- Voyager Therapeutics VY-HTT01 (HTT, AAV1.miHTT) — currently in early clinical evaluation. AAV gene therapy has its own unique toxicity profile (DRG toxicity, hepatotoxicity at high IV doses, complement activation, integration concerns) that is largely orthogonal to small-oligonucleotide toxicity.
- uniQure AMT-130 (HTT, AAV5-miHTT) intrastriatal — Phase 1/2 ongoing.

7.6 New chemistries on the horizon

- **5′-CP DNA** (Obika/Yamaguchi 2025) — clinical evaluation pending.
- **PN-backbone (phosphoryl-guanidine, Wave)** — WVE-003 is the first clinical proof-of-concept.
- **Triantenary GalNAc-LICA optimizations** continue to reduce systemic doses required.
- **CRISPR-Cas-based CNS therapeutics** (e.g., NTLA-2001 amyloidosis) overlap with oligonucleotide field but introduce distinct toxicity (guide RNA innate immunity, AAV delivery liabilities).

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8. Key Research Groups & Companies

Academic labs (selected)

- **Anastasia Khvorova / Neil Aronin (UMass Chan)** — di-siRNA chemistry, divalent cation toxicity work, allele-selective HTT siRNA. Spinout: Atalanta Therapeutics.
- **Jonathan Watts (UMass Chan)** — PS-content and buffer effects on CNS toxicity (Moazami 2024).
- **Stanley T. Crooke (Ionis emeritus; n-Lorem Foundation)** — paraspeckle protein binding, Gap2-OMe, 2′-F protein degradation.
- **Adrian Krainer (CSHL)** — splice-switching ASO scientific founder, nusinersen mechanism.
- **Don Cleveland (UCSD)** — original SOD1 ASO concept; FUS biology.
- **Timothy Miller (Washington University)** — tofersen clinical lead investigator.
- **Sarah Tabrizi (UCL)** — tominersen clinical lead investigator.
- **Neil Shneider (Columbia)** — jacifusen/ION363 compassionate-use lead.

- **Takanori Yokota (TMDU/Tokyo)** — intracellular Ca^{2+} mechanism, G-clamp/aminoethoxyphenoxazine.
- **Satoshi Obika / Takao Yamaguchi (Osaka University)** — 5′-CP late-onset toxicity rescue.
- **Peter H. Hagedorn (Roche Copenhagen, ex-BMS)** — sequence-based toxicity prediction, hepatotoxicity ML.
- **Matthew Wood (Oxford)** — peptide-conjugated PMOs.
- **Frank Bennett & Brett Monia (Ionis)** — long-time ASO chief scientists; tofersen, BIIB080 chemistry.

Companies

Company	Platform	CNS programs (selected)
Ionis Pharmaceuticals	2′-MOE/PS gapmer, GalNAc-LICA	Tofersen, BIIB080, ION363, ION582
Biogen	Partner / commercializer	Spinraza, Qalsody, BIIB080 (CELIA), Zorevunersen
Alnylam	Fully-modified siRNA, GalNAc	Vutrisiran (peripheral); CNS via partnerships
Roche / Genentech	ASO + small molecules	Tominersen, Atalanta partnership
Wave Life Sciences	Stereopure PS, PN backbone	WVE-003 (HTT SNP3), WVE-N531 (DMD)
Stoke Therapeutics	TANGO splice ASO	Zorevunersen (Dravet)
Atalanta Therapeutics	di-siRNA	ATL-201 (KCNT1), ATL-101 (HTT)
Avidity Biosciences	AOC (anti-TfR1)	AOC 1001 (DM1), AOC 1044 (DMD exon-44), AOC 1020 (FSHD)
Dyne Therapeutics	FORCE Fab-AOC	DYNE-101 (DM1), DYNE-251 (DMD)
PepGen	Cell-penetrating peptide-PMO	PGN-EDO51 (DMD), PGN-EDODM1 (DM1)

Denali Therapeutics	ETV TfR1 shuttle	DNL310 (Hunter), oligonucleotide programs
Voyager Therapeutics	AAV-shRNA	VY-HTT01
uniQure	AAV-miRNA	AMT-130 (HTT)
n-Lorem Foundation	Personalized ASOs (PS-MOE)	“N-of-1” treatments — institutional focus on CNS toxicity given small/no preclinical animal work

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9. Data Sources & Where to Find More Information

Journals (prioritized)

1. **Molecular Therapy** (Cell Press) — flagship for nucleic acid therapeutics
2. **Molecular Therapy — Nucleic Acids** (open access) — core methodology / safety papers
3. **Nucleic Acids Research** (Oxford, open access) — chemistry / biology mechanism
4. **Nature Biotechnology** — landmark chemistry milestones
5. **Nature Medicine, NEJM, Lancet Neurology** — pivotal clinical trial results
6. **Nature Reviews Drug Discovery** — annual oligonucleotide pipeline reviews (Roberts, Langer & Wood)
7. **Annual Review of Pharmacology and Toxicology** — Crooke chapters
8. **Nucleic Acid Therapeutics** (Mary Ann Liebert, OTS journal) — community-of-practice methods
9. **JCI Insight, Brain Communications** — translational CNS work

Conferences

- **OTS (Oligonucleotide Therapeutics Society)** — annual fall meeting; the field’s flagship gathering.
- **ASGCT (American Society of Gene and Cell Therapy)** — May annually; large clinical-trial readouts.
- **AAN, ANA, ECCM, MDS Congress** — for clinical neurology of HD, ALS, AD, Dravet.
- **TIDES (USA + Europe)** — industrial CMC / development focus.
- **AD/PD International Conference** — for AD ASO updates.

Preprint servers

- **bioRxiv** (especially q-bio.MN, q-bio.QM tags) — Khvorova, Watts, Hagedorn groups regularly post here months before publication.
- **chemRxiv** — emerging chemistry (e.g., the 2025 ML "sequence-chemistry collusion" paper).

Regulatory and clinical databases

- **ClinicalTrials.gov** — NCT numbers cited above.
- **FDA accessdata.fda.gov** — full labels, integrated safety reviews, AdComm briefings (Spinraza 209531, Qalsody 215887).
- **EMA EPAR documents** at ema.europa.eu.
- **Drugs@FDA** for "AppLetter" + medical/clinical pharmacology reviews.

Industry sources

- Ionis, Alnylam, Biogen, Roche, Wave, Stoke investor decks and pipeline updates.
- Atalanta Therapeutics press releases (atalantatx.com).
- HDBuzz (en.hdbuzz.net) — accessible coverage of HD clinical trials.
- CureFFI.org (Eric Vallabh Minikel) — sharp prion/ASO trial commentary.

Patent databases

- Google Patents and USPTO for "phosphorothioate antisense oligonucleotide" + relevant company assignees.
- Key foundational patents: Crooke/Ionis PS-MOE platform; Khvorova/UMass di-siRNA; Wave stereopure synthesis; Stoke TANGO platform.

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10. Knowledge Taxonomy / Structure of the Field

A practical mental map of CNS oligonucleotide toxicity organizes along **four orthogonal axes**:

Axis 1: Time course

- **Acute (seconds–hours)** → divalent cation chelation, sequence-dependent seizures, motor phenotypes
- **Subacute (days–weeks)** → CSF inflammation, NfL rise, microglia activation, ventricular volume change

- **Late-onset (days–weeks–months)** → paraspeckle/nucleolar stress → p53 → apoptosis (Yamaguchi 2025: hypoactivity persisting for six days or more after dosing)
- **Chronic (months–years)** → cumulative on-target depletion effects (e.g., total HTT lowering)

Axis 2: Mechanism dependency

- **Hybridization-dependent** → RNase H off-target cleavage; mis-splicing; on-target over-suppression
- **Hybridization-independent** → PS-protein binding (paraspeckle, RNase H1, hnRNPs); cation chelation; TLR9 sensing of CpG

Axis 3: Source of variability

- **Sequence-dependent** → G-richness, CpG content, off-target match score
- **Chemistry-dependent** → PS content, sugar modification (LNA > cEt > MOE for hepatotoxicity; 2′-F → P54nrb degradation; 5′-CP rescues late-onset; PMO benign)
- **Formulation-dependent** → divalent cation content, buffer choice, particle formation
- **Delivery-dependent** → IT vs ICV; volume; rate; device

Axis 4: Cellular target of damage

- **Neurons** (acute Ca^{2+} disruption; late-onset apoptosis)
- **Microglia/astrocytes** (TLR9/inflammatory activation)
- **Choroid plexus / ependyma** (CSF absorption, ventricular volume)
- **Vascular/meningeal** (aseptic meningitis, intracranial hypertension)
- **DRG / peripheral nerve** (branaplam-type)

A toxicity finding should always be located on all four axes — this is how Roche’s tominersen failure resists simple explanation: late-subacute, partially hybridization-independent (PS/protein), highly chemistry-dependent (full-PS gapmer), and partly hybridization-dependent (on-target total HTT lowering), affecting neurons, microglia, and CSF dynamics simultaneously.

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11. Key Papers to Read First (in this order)

1. **Khvorova A, Watts JK. “The chemical evolution of oligonucleotide therapies of clinical utility.” Nat Biotechnol 2017;35:238-248.** — the foundational chemistry primer.

2. **Crooke ST. "Molecular Mechanisms of Antisense Oligonucleotides." Nucleic Acid Ther 2017;27:70-77.** — the canonical mechanism review.
3. **Hagedorn PH et al. "Acute Neurotoxicity of Antisense Oligonucleotides After Intracerebroventricular Injection Into Mouse Brain Can Be Predicted from Sequence Features." Nucleic Acid Ther 2022;32:151-162.** DOI 10.1089/nat.2021.0071, PMC9221153.
4. **Miller R, Paquette J, Barker A et al. (Khvorova/Aronin). "Preventing acute neurotoxicity of CNS therapeutic oligonucleotides with the addition of Ca^{2+} and Mg^{2+} in the formulation." Mol Ther Nucleic Acids 2024;35:102359.** PMC11185713.
5. **Moazami MP, Rembetsy-Brown JM, Sarli SL et al. (Watts). "Quantifying and mitigating motor phenotypes induced by antisense oligonucleotides in the central nervous system." Mol Ther 2024.** PMC11638874.
6. **Jia C, Lei Mon SS, Yang Y et al. (Yokota). "Change of intracellular calcium level causes acute neurotoxicity by antisense oligonucleotides via CSF route." Mol Ther Nucleic Acids 2023;31:182-196.** PMC9843489.
7. **Yamaguchi T et al. (Obika). "Unraveling and controlling late-onset neurotoxicity of antisense oligonucleotides through strategic chemical modifications." Mol Ther Nucleic Acids 2025;33:102692.** PMC12744863. *(The 5′-CP paper.)*
8. **Vickers TA, Rahdar M, Prakash TP, Crooke ST. "Kinetic and subcellular analysis of PS-ASO/protein interactions with P54nrb and RNase H1." Nucleic Acids Res 2019;47:10865-10880.** PMC6846478.
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12. Recommendations (decision-ready)

For a preclinical CNS-ASO/siRNA design team:

1. **Adopt the Khvorova formulation rule** — every preclinical and clinical CNS oligonucleotide formulation should contain Ca^{2+} + Mg^{2+} at molar excess of PO+PS bonds in HEPES or lactate buffer; do not use phosphate-based buffers with Ca^{2+}-containing oligos. **Threshold to escalate**: if any seizure event is observed in mouse ICV, reformulate before redesigning sequence.
2. **Pre-screen 50–100 candidate sequences by the Hagedorn calcium-oscillation assay** before any in vivo work; eliminate the top tertile by 3′-G-content and CpG content. **Threshold to escalate**: any in vivo behavioral score >2 on the 0–3 Hagedorn scale.
3. **Adopt mixed PS/PO backbones with MOE wings as default**; reserve LNA/cEt for rare cases requiring extreme affinity; **never use 2′-F in gap region**.
4. **Add the 5′-CP modification at gap positions** for late-onset toxicity mitigation if the program will involve chronic CNS dosing. **Threshold to escalate**: any mouse ICV behavioral phenotype at day 7–14 post-dose.
5. **Conduct a Gap2-OMe walk** as a standard medicinal chemistry maneuver to reduce hybridization-independent protein binding.
6. **Adopt the Moazami extended behavioral scoring and Watts awake-EEG/EMG protocols** as in vivo standards.
7. **Plan NHP IT toxicology with serial CSF NfL/GFAP/cell count/protein** as primary safety biomarkers, not just histopathology.

For a clinical development team:

1. **Use NfL as a co-primary safety biomarker** from first-in-human; pre-specify trajectory stopping rules (e.g., >2-fold sustained CSF NfL rise over 3 consecutive timepoints triggers iDMC review).
2. **Use MRI ventricular volume** as a secondary structural safety endpoint with ≥annual cadence in any chronic CNS program.

3. **CSF cell count + protein at every CSF sampling** — pleocytosis or protein elevation >50 mg/dL above baseline is a sentinel signal (tofersen and zorevunersen precedents).
4. **Use Q16W as starting dose interval for gapmers** — Q8W is unsafe at high doses (tominersen precedent).
5. **For adult/older pediatric programs, use ultrasound-guided lumbar puncture** to minimize procedural AEs.
6. **For genetically defined fatal CNS diseases, the FDA accelerated-approval-with-NfL pathway (tofersen precedent) is the explicit target** — design Phase 2 to deliver registration-quality NfL data.

For investors / strategy:

1. **Di-siRNA (Atalanta) and AOC-CNS (Avidity, Dyne, Denali) are the two leading next-generation delivery platforms** to monitor for first-in-class clinical proof-of-concept in 2025–2027.
2. **Genetic medicines for HD remain a high-risk-high-return area**; positive WVE-003 SELECT-HD signals and GENERATION HD2 100-mg interim safety reset the field, but the burden of proof on safety is now extreme.
3. **TANGO splice-upregulation (Stoke) is the most novel mechanism with clinical validation outside of nusinersen-style splice modulation** — watch zorevunersen EMPEROR readout (mid-2027).

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13. Caveats

1. **Tominersen mechanism remains unsettled.** The NfL/ventricular volume profile is a composite of chemistry-driven inflammation, on-target total HTT lowering, and possibly individual susceptibility — neither Roche nor the field has cleanly deconvoluted these.
2. **The 7% serious neurologic AE rate of tofersen** comes from a small denominator (~145 patients at 100 mg) and the natural history of ALS confounds attribution; the comparison to placebo is suggestive but not statistically clean.
3. **The Yamaguchi 5′-CP paper (2025) is recent and not yet validated by independent labs at clinical scale**; the late-onset mouse model itself is non-standard.
4. **The Khvorova divalent-cation rescue, while reproduced in multiple species, has not yet been used in a published clinical formulation.** Spinraza, Qalsody, and Tominersen formulations rely on artificial CSF compositions with physiological calcium; whether modifying them to include excess calcium would alter clinical safety is untested.

5. **Quantitative figures for di-siRNAs (Atalanta) in humans do not exist yet** — IND submissions planned for 2025; the platform's safety profile in humans is unknown.

6. **The peripheral neuropathy of branaplam is mechanistically distinct from oligonucleotide toxicity** (small-molecule splice modulator) but converges on a shared nucleolar-stress/p53 pathway — generalization should be cautious.

7. **The "platform toxicology" approach assumes chemistry uniformity** — for novel chemistries (5′-CP, PN backbone, di-siRNA), full safety pharmacology and chronic NHP studies are still required.

8. **Conflicting numerical reports**: zorevunersen seizure reductions vary by source (85%/74% at 3/6 mo for 70 mg in early press; 58–90% median over OLE up to 36 mo per NEJM 2026). The discrepancy reflects pooling of cohorts and timepoints; primary NEJM publication (Laux et al. 2026) should be cited preferentially.

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End of report.